Adverse Drug Reaction Case Analysis

Suspected co-trimoxazole-induced maculopapular rash

1. Case Summary

Patient	62-year-old man, 70 kg
Comorbidities	Type 2 diabetes, hypertension, dyslipidemia
Allergies / organ function	No known drug allergies. Normal renal and hepatic function
Regular drugs	Metformin 500 mg BD (5 yrs); amlodipine 5 mg OD (3 yrs); atorvastatin 20 mg OD (2 yrs); enalapril 5 mg OD (6 months)
New drug	Co-trimoxazole DS BD for UTI, started Day 0
Event	Day 8: itchy, generalized maculopapular rash with low-grade fever (99.5°F). No mucosal involvement, blisters, or facial swelling
Investigations	Eosinophils 8%; LFTs and creatinine normal; viral screen negative
Outcome	Co-trimoxazole stopped Day 9; cetirizine given; rash resolved by Day 14. No rechallenge

2. Timeline

Day	Event
Months to years before	Metformin, amlodipine, atorvastatin, enalapril taken without problems
Day 0	Co-trimoxazole DS BD started
Day 8	Pruritic maculopapular rash and low-grade fever appear
Day 9	Co-trimoxazole stopped (dechallenge); cetirizine started
Day 14	Rash resolved

An onset 8 days after first exposure fits a delayed (Type IV) hypersensitivity reaction, which typically appears 7-14 days after starting a drug.

3. Suspect Drug

Co-trimoxazole (the sulfamethoxazole component is the usual culprit).

- Well-documented cause of cutaneous reactions.
- The only new drug introduced before the event.
- Rash resolved after withdrawal (positive dechallenge).
- Enalapril, amlodipine, atorvastatin, and metformin are unlikely: long stable use and no new exposure before onset.

4. Causality Assessment

Naranjo scale: total score 7 (Probable)

Question	Answer	Score
Previous conclusive reports of this reaction?	Yes	+1
Did the event follow the drug?	Yes	+2
Improved on stopping the drug?	Yes	+1
Reappeared on rechallenge?	Not done	0

Question	Answer	Score
Alternative causes?	No (viral screen negative, other drugs long-tolerated)	+2
Placebo reaction, drug levels, dose-response, prior similar reaction	Not applicable / unknown	0
Confirmed by objective evidence?	Yes (eosinophilia)	+1
Total		**7**

Naranjo interpretation: 9+ definite, 5-8 probable, 1-4 possible, 0 or less doubtful.

WHO-UMC: Probable / Likely. The timing is reasonable, other causes are unlikely, and the response to withdrawal is clinically reasonable. Rechallenge was not performed (and should not be, given the risk).

5. Classification

Parameter	Finding	Rationale
Type	Type B (bizarre)	Not dose-related; delayed hypersensitivity (Type IV)
Severity (Hartwig)	Level 3 (mild to moderate)	Drug stopped and antihistamine needed; no prolonged hospital stay
Preventability (Schumock-Thornton)	Not preventable	Appropriate drug and dose; no prior allergy history

6. Management and Reporting

- Stop co-trimoxazole. Choose an alternative UTI antibiotic based on culture (for example nitrofurantoin or cephalexin, depending on sensitivity).
- Record "sulfonamide/co-trimoxazole: rash" in the allergy history and counsel the patient to avoid it.
- Monitor for red flags: mucosal lesions, blistering, skin pain, or organ involvement (SJS/TEN, DRESS).
- Report to the Pharmacovigilance Programme of India (PvPI) using the suspected ADR reporting form.

7. Teaching Point

Co-trimoxazole combined with an ACE inhibitor (enalapril) raises the risk of **hyperkalemia**. This combination should prompt a serum potassium check, even though it did not cause this reaction.